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A hybrid model integrating RoBERTa, TF-IDF, and attention mechanism for medical query intent classification

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In medical retrieval scenarios, accurate identification of user query intent is pivotal for enhancing retrieval relevance and knowledge augmentation effectiveness. However, existing approaches exhibit polarized performance: traditional shallow models (e.g., TF-IDF, Bag-of-Words) struggle to capture contextual semantics, while pre-trained language models, despite their strong deep semantic representation capabilities, often overlook keyword weighting and category-specific priors. This leads to suboptimal accuracy for long-tail categories and susceptibility to confusion from emotional or colloquial query formulations. To address these challenges, this study proposes a hybrid model integrating RoBERTa-wwm-ext with sentence-level TF-IDF and category-centroid TF-IDF via a three-branch attention fusion mechanism. The model employs adaptive attention mechanisms to dynamically weight deep-shallow features at the sample level, while incorporating category-centroid vectors as explicit semantic anchors to mitigate class imbalance issues. Experimental evaluations on the KUAKE-QIC dataset demonstrate that our approach achieves an accuracy of 0.824 and a Macro-F1 score of 0.800, outperforming all baseline models by 2.3% points. These results highlight the synergistic benefits of deep-shallow feature interaction, ensuring robust recall for dominant categories while significantly improving precision for minority classes. This work provides a generalizable framework for medical intent classification that balances semantic integrity with keyword-driven priors, laying a more reliable foundation for applications such as clinical question-answering systems, medical search ranking, and intelligent triage systems.

Keywords Medial intent classification, RoBERTa, TF-IDF, Attention mechanism, Knowledge-enhanced retrieval

In the context of the internet era, the accuracy of medical text intent classification holds paramount importance for patient consultation and diagnosis. Surveys indicate that up to 80% of internet users have engaged in online searches for medical and health-related information¹. However, due to the professional complexity of medical knowledge, accurately classifying such queries remains a significant challenge. Medical search queries are often extremely concise, semantically ambiguous, and may use non-professional terminology to describe symptoms or conditions^{2,3}. For instance, a single symptom may have multiple colloquial names, complicating query comprehension. Furthermore, incorrect intent judgment risks providing misleading medical information, which could adversely impact user decision-making or even delay medical care⁴. Consequently, accurately understanding the underlying intent behind users' medical queries has emerged as an urgent problem in the field of medical information retrieval.

Medical text processing encounters numerous unique challenges. First, medical texts exhibit strong domain-specific characteristics, and direct application of generic natural language processing (NLP) models to medical corpora often yields suboptimal performance. This is primarily attributed to significant lexical distribution and semantic discrepancies between medical and general-purpose corpora, leading to a lexical shift issue. For instance, studies have shown that general-purpose pre-trained models like BERT underperform on biomedical texts unless further pre-trained on large-scale medical corpora⁵. Second, medical texts—including patient-reported symptoms and electronic health records—are frequently unstructured and noisy, with patients often

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using imprecise or ambiguous colloquial language³. This necessitates models capable of incorporating medical domain knowledge for contextual interpretation, such as mapping patient terminology to standardized medical concepts. While prior research has attempted to extract medical entities or leverage medical knowledge graphs for enhanced representation, ineffective integration risks information loss or excessive model complexity². Third, annotated data scarcity remains a critical bottleneck. High-quality training data for medical intent classification requires expert annotation, which is costly and limited in scale, resulting in insufficient learning of rare terms and infrequent intent categories⁶. Category imbalance is particularly pronounced in medical queries, where specialized intents (e.g., rare disease consultations) are underrepresented, causing models to bias toward majority classes. Addressing these challenges requires solutions for lexical semantic adaptation, short-text context enrichment, domain knowledge fusion, and mitigation of data sparsity and imbalance⁷.

This paper proposes a principled hybrid model that fuses (i) contextual representations from RoBERTa-wwm-ext, (ii) sentence-level TF-IDF features, and (iii) class-prototype (centroid) TF-IDF priors via an additive attention module. The attention layer adaptively weights branches per input, enhancing discrimination for minority and boundary-ambiguous intents. To prevent label leakage during inference, we adopt soft-prototype inference, forming input-dependent prototype expectations from model-predicted distributions. Our main contributions are as follows:

- (1) Hybrid multi-branch framework for medical query intent classification. We design a three-branch model that integrates deep semantic features from RoBERTa-wwm-ext with two complementary shallow representations — sentence-level TF-IDF for local keyword salience and category-centroid TF-IDF for global class priors. This combination enhances both contextual understanding and category discrimination for ultra-short medical queries.
- (2) We introduce an inter-branch attention mechanism that dynamically adjusts the contribution of each feature stream based on query content, rather than using static concatenation. This adaptive weighting improves precision for minority classes while maintaining robust recall for dominant ones.
- (3) Validated effectiveness and practicality. On KUAKE-QIC, the model attains 0.824 accuracy/0.800 macro-F1, a +2.3 pp macro-F1 gain over a strong RoBERTa baseline, supported by ablations (branch removal; attention vs. concatenation/gating/late fusion), sensitivity studies (projection dimension, TF-IDF feature count), error analysis on low-performing categories, and deployment-oriented efficiency metrics.

Related work

The evolution of medical text classification has witnessed a paradigm shift from traditional methods to deep learning techniques. Early approaches primarily relied on rule-based systems or classical machine learning algorithms. Vector space models, leveraging term frequency-inverse document frequency (TF-IDF) and bag-of-words (BoW) representations, were initially employed to capture lexical significance⁸. Subsequently, word embedding technologies such as Word2Vec⁹ and GloVe¹⁰ emerged, enabling semantic relationships between terms to be modeled through continuous vector spaces. Faris et al.¹¹ demonstrated the utility of particle swarm optimization for feature selection combined with a one-vs-rest SVM ensemble, achieving patient department classification. While these traditional methods performed adequately on small-scale datasets, they heavily depended on expert-driven feature engineering and struggled to capture deep contextual semantics. In recent years, deep learning has revolutionized medical text processing. Architectures such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs)/long short-term memory (LSTM) networks have enabled automatic feature learning from raw data, reducing reliance on manual design³. For instance, Du et al.¹² proposed the ML-Net model, which employs deep neural networks for multi-label classification of biomedical literature, significantly enhancing efficiency over traditional approaches. Chen et al.¹³ integrated LSTM networks into medical text classification frameworks, improving the extraction of critical symptom keywords and descriptive elements from large-scale medical corpora. Zhang et al.¹⁴ explored CNN architectures for text classification, demonstrating how convolutional operations and max-pooling layers capture local textual features to achieve efficient categorization. Liu et al.¹⁵ introduced a recursive neural network (RNN) framework combining online language understanding and language modeling, showcasing superior performance on context-dependent tasks. The advent of attention mechanisms, pioneered by Bahdanau et al.¹⁶, further advanced model capabilities, yielding improved performance in machine translation and text classification. Chai et al.¹⁷ leveraged statistical text classification to identify medical IT incidents, achieving notable accuracy gains. Ali et al.¹⁸ and Ozyegen et al.¹⁹ separately proposed machine learning-based and word-level text highlighting approaches for medical text classification, enhancing clinical data analysis efficiency. More recently, Liang et al.²⁰ introduced a medical text classification framework combining ZEN and capsule networks, while Luo et al.²¹ demonstrated the efficacy of the bilingual Taiyi model for diverse biomedical tasks. These advancements collectively underscore the transformative potential of deep learning and hybrid architectures in addressing the unique challenges of medical text processing.

This progress has catalyzed the development of domain-adapted BERT variants, including BioBERT²⁴ for biomedical texts, and Chinese-specific models such as MC-BERT²⁵, BERT-Healthcare²⁶, and ClinicalBERT²⁷. Innovative hybrid architectures further extend these capabilities: Li et al.²⁸ proposed ECC-BERT, while Wa et al.²⁹ and Yu et al.³⁰ introduced BERT-CNN and TinyBERT-CNN fusion models, respectively, combining BERT's semantic robustness with convolutional neural networks (CNNs) to enhance classification performance. Additionally, Wang et al.³¹ demonstrated the efficacy of discriminative pre-training and prompt tuning techniques, and Peng et al.³² showcased transfer learning approaches for medical text classification. Zhang et al.³³ highlighted BERT's applicability in biological sequence analysis, achieving notable results in predicting lncRNA subcellular localization. By incorporating large-scale medical literature and clinical note corpora during pre-training, these models demonstrate superior adaptation to medical terminology and discourse patterns.

They achieve state-of-the-art performance across medical named entity recognition, relation extraction, and user intent classification tasks. BERT-based deep language models have emerged as cornerstone technologies in contemporary medical NLP, providing critical tools for complex semantic interpretation and intent recognition in clinical settings.

Despite the remarkable performance of deep learning models like BERT in medical search intent classification, a growing research trend emphasizes hybrid architectures that integrate multi-source features. These approaches synergize deep semantic representations with traditional statistical features and domain-specific knowledge to leverage their complementary strengths. Statistical features such as TF-IDF provide intuitive measures of term significance, compensating for deep models' potential oversight of rare but critical medical terminology³⁴. Recent studies demonstrate the efficacy of incorporating TF-IDF-based gating mechanisms within RoBERTa models to downweight less informative terms while emphasizing emotionally salient or symptom-related keywords, thereby improving classification accuracy³⁵. In the medical domain, Mao et al.¹¹ proposed a Hybrid Model that fuses traditional TF-IDF features with BERT-derived semantic vectors and features extracted via LSTM/CNN architectures. Through an attention-based decision fusion mechanism, this model achieves high-accuracy classification of patient symptom descriptions, exceeding 93.5% accuracy on a Chinese symptom dataset spanning eight clinical departments and outperforming traditional standalone methods. Zhang et al.³⁶ introduced the E-BERT architecture, which integrates medical entity embeddings and entity-aware attention mechanisms into the BERT framework. By explicitly encoding medical entities (e.g., symptoms, anatomical locations, diseases) and their relationships, this model demonstrates superior performance on lengthy clinical text classification tasks. Other approaches employ reinforcement learning to dynamically identify discriminative features. For instance, Nie et al.³⁷ developed a keyword-driven reinforcement learning model for Chinese respiratory disease intent classification, using term-frequency reward functions to select critical information words. These words are then fed into LSTM and graph convolutional networks, enhancing intent recognition performance. To address short-text sparsity, Duan et al.³⁸ proposed a TF-IWF-based keyword weighting scheme combined with Word2Vec synonym expansion, effectively mitigating feature sparsity and semantic ambiguity. Collectively, hybrid models that fuse traditional features (keyword frequencies, TF-IDF), domain knowledge (medical entities, knowledge graphs), and deep contextual representations now represent the forefront of medical search intent classification research^{39–42}. These architectures utilize attention mechanisms as dynamic “glues” to balance information weights across modalities, achieving superior performance and robustness compared to single-modality approaches. This fusion strategy underscores a paradigm shift toward knowledge-augmented, context-aware systems in medical NLP applications.

Previous studies primarily focus on either deep models (e.g., BERT, RoBERTa) or statistical keyword features (e.g., TF-IDF, LDA). Deep models capture context but often overlook rare, domain-specific medical terms, while shallow models emphasize word importance yet lack contextual understanding. To bridge this gap, our work proposes a three-branch hybrid architecture combining contextual embeddings (RoBERTa-wwm-ext), local keyword salience (sentence-level TF-IDF), and global category priors (category-centroid TF-IDF). This design aims to achieve both semantic precision and category-level robustness for ultra-short medical queries.

Methodology

Model framework

This paper proposes a multi-feature fusion hybrid model, whose architecture is structured into three hierarchical components from top to bottom as shown in Fig. 1: the Feature Generation Layer, responsible for extracting semantic information and lexical features from medical queries; the Feature Fusion Layer, which integrates cross-modal information through adaptive weighting to dynamically balance semantic and lexical representations; and the Classification Decision Layer, performing final intent classification using fused features for discriminative multi-class prediction.

The first component of the model is feature generation, whose core objective is to construct complementary information representations for the same query. The RoBERTa branch contributes long-range contextualized deep semantic vectors, the sentence-level TF-IDF branch highlights the most discriminative medical keywords within the current sentence for intent discrimination, while the category-centric TF-IDF branch incorporates global category priors derived from training corpus statistics. This integration effectively mitigates data sparsity issues arising from long-tail categories and rare terminology by encoding these priors into the representation space. All three feature streams are linearly projected to identical dimensions, establishing a unified dimensionality basis for subsequent weighted fusion operations, as detailed in the Feature Generation Layer of Table 1.

The second component of the model is attentive fusion, which dynamically determines the contribution ratio of the three feature branches based on query-specific content. Through learnable nonlinear scoring functions and softmax normalization, the model automatically amplifies the feature branch that best explains the current query intent. When queries contain highly indicative medical terminology, the TF-IDF branch receives greater emphasis. For semantic dependencies requiring long-range contextual understanding, the RoBERTa branch becomes more dominant. In scenarios with extreme class imbalance, the category centroid branch provides supplementary support for underrepresented classes. The resultant fused representation integrates contextual semantics, keyword salience, and categorical knowledge, demonstrating enhanced robustness against noise and data distribution skew.

Inter-branch attention vs. intra-sequence self-attention. RoBERTa's self-attention operates within a sequence to model token-token dependencies and form contextual representations. Our fusion module instead attends across branches over three heterogeneous features (RoBERTa embedding, sentence-level TF-IDF, category-centroid TF-IDF). It acts as a sample-wise gating mechanism to adaptively weight each branch for a specific query, rather than recomputing token-level dependencies as in Transformer self-attention.

Let the three features be f_1 (Sentence TF-IDF), f_2 (RoBERTa-wwm-ext), and f_3 (Class TF-IDF Centroid).

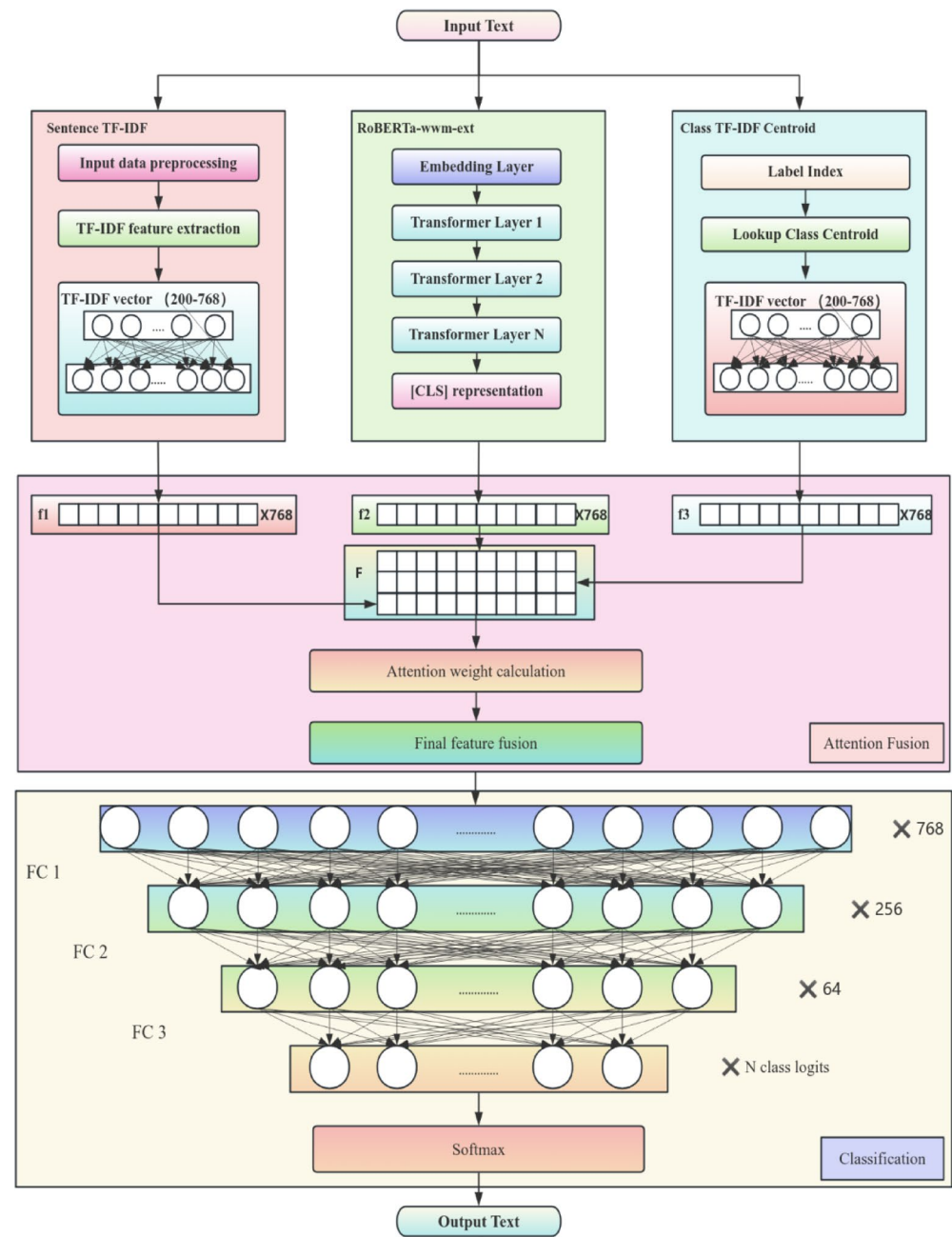


Fig. 1. Model Framework Diagram.

Branch	Principal components	Output dimension	Functional role
Sentence TF-IDF	1. Tokenization 2. TF-IDF Vectorizer(max_features = 200)	200 → 768	Capturing significance of intra-query keywords through sparse high-dimensional encoding
RoBERTa-wwm-ext	1. Embedding Layer 2. 12× Transformer Layers 3. [CLS] Pooling	768	Generating context-aware semantic representations with whole-word masking pre-training
Class TF-IDF Centroid	1. Centroid vector retrieval via category lookup table 2. Linear projection	200 → 768	Encoding global category priors and domain-specific lexical distributions

Table 1. Feature generation Component.

1. Stack the three 768 dimensional representations into a matrix by line, as shown in Formula (1).

$$F = \begin{bmatrix} f_1^\top \\ f_2^\top \\ f_3^\top \end{bmatrix}, \quad d = 768 \tag{1}$$

2. The calculation of attention weight is shown in Formula (2).

$$\alpha_i = \text{Softmax} \left(u^\top \tanh(W f_i) \right), \quad i = 1, 2, 3, \tag{2}$$

Where, W, u are learnable parameters.

3. Feature aggregation, as shown in Formula (3).

$$f_{fused} = \sum_i i = 1^3 \alpha_i f_i \in \mathbb{R}^{768} \tag{3}$$

Where, α_i represents the importance coefficient of the three-way feature. For queries with key information words, TF-IDF branch gets higher weight; RoBERTa is more important for questions with strong context dependence. So as to realize dynamic weighting.

The third component is classification decision-making, which functions to transform the fused high-dimensional features into specific intent predictions. Utilizing a two-stage dimensionality-reducing fully connected network (768→256→64) combined with ReLU activation and Dropout regularization, the model incorporates nonlinear discriminative capabilities and prevents overfitting, enabling learning of complex feature combination patterns. The final layer maps the 64-dimensional representation to 11-dimensional logits, followed by a Softmax operation to output probability distributions across intent categories, achieving final multi-class classification as detailed in Table 2.

RoBERTa-based semantic encoder

RoBERTa is a pre training language model based on Transformer architecture. Transformer is a neural network architecture completely based on self attention mechanism, including encoder and decoder.

the Transformer architecture is alternately stacked by multi head self attention and feedforward networks. BERT and RoBERTa are variants of its Encoder. Each layer first executes Multi-Head Self-Attention extract intra sentence global dependencies, and then provides nonlinear mapping through Add&Norm and Feed Forward. Residual connection and layer normalization jointly stabilize the deep gradient. Compared with RNN, Transformer has the advantages of parallel computing and long-distance modeling. Compared with CNN, the attention mechanism can dynamically focus on any position, eliminating the limitation of fixed receptive field.

RoBERTa introduces three key enhancements over the original BERT training paradigm: (1) Removal of Next-Sentence Prediction to eliminate task-related noise. (2) Adoption of dynamic masking, where 15% of tokens are randomly re-masked in each training epoch. (3) Utilization of larger mini-batch sizes and extended training steps. These optimizations significantly improve representation quality and transfer robustness. For Chinese applications, the HFL team from Harbin Institute of Technology conducted continued pre-training on 160 GB of Chinese encyclopedic, news, Q&A, and forum data, developing RoBERTa-wwm-ext. Its critical innovation lies in Whole-Word Masking, which uniformly masks all subword components of Chinese terms (e.g., treating “甲状腺功能减退” as a complete medical compound rather than fragmented “甲状 + 腺 + 功能...”). This study employs this variant as the semantic encoder, with selection rationale detailed in Table 3.

The internal computer management of RoBERTa-wwm-ext in the coding phase is shown below.

1. RoBERTa Encoder receives the token sequence $\{x_1, x_2, x_3, \dots, x_L\}$ with a length of $L \leq 256$. First, add the token, position and segment information to get the input embedding, as shown in Formula (4).

$$e_i = E_{\text{tok}}(x_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0), \quad i = 1 \dots L. \tag{4}$$

Where, x_i represents the i th sub word token, E_{tok} represents word vector, E_{pos} represents position vector, E_{seg} represents segment vector, E_i indicates the embedding of input to the i th layer Transformer.

2. For the calculation of multi head attention, for the first layer, the query, key and value are obtained by linear mapping. See Formula (5) for details.

$$Q = H^{(l-1)} W_Q, \quad K = H^{(l-1)} W_K, \quad V = H^{(l-1)} W_V. \tag{5}$$

Layer	Input Dimension	Output Dimension	Functional Purpose
FC 1	768	256	Dimensionality reduction and extraction of high-order combinatorial features
FC 2	256	64	Introduction of nonlinear discriminative capabilities through activation functions
FC 3	64	N (N=11)	Generation of N=11-dimensional logits for multi-class probability distribution

Table 2. Classification decision Component.

Model	BERT-base-Chinese	RoBERTa-base-Chinese	RoBERTa-wwm-ext	Advantage analysis
Pre training corpus size	≈ 3 GB Wiki	≈ 5 GB Wiki + News	≈ 5.4 GB Wiki + News + QA + Forums	More abundant corpus, covering oral symptom expression
Mask mode	Sub word level (Static)	Sub word level (Dynamic)	Whole-Word (Dynamic)	Fully learn the overall semantics of medical compounds
NSP task	Include	Removal	Removal	Reduce noise and focus on language modeling
Batch/Steps	256/1 M	8 K/500 K	8 K/1 M	More training and generalization
Parameter quantity	102 M	102 M	110 M	The structure is the same, and the word list and Position Embedding are added
Public performance of Chinese tasks	baseline	+ 1.1 pp F1	+ 2 pp F1	Best performance, improved long tail word recall

Table 3. Comparison of semantic encoder Models.

Where, $H^{(l-1)}$ is the output of layer $l-1$, and W_Q , W_K , and W_V are learnable weights. Q, K, V are query, key and value matrices.

Then calculate the scale point product attention, as shown in Formula (6).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}. \quad (6)$$

Where, $d_k=64$ is a single header dimension. The multi head mechanism parallelizes $h=12$ subspaces. Then conduct multi head splicing, as shown in Formula (7).

$$\mathbf{Z} = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}_O, \text{head}_j = \text{Attention}(\mathbf{Q}_j, \mathbf{K}_j, \mathbf{V}_j). \quad (7)$$

Where, h represents the number of heads, W_O represents the output linear mapping, and Z represents the attention layer output.

3. Feed forward layer and residual. The output of each layer is normalized by residual and LayerNorm, as shown in Formula (8).

$$\tilde{\mathbf{H}}^{(l)} = \text{LayerNorm}(\mathbf{H}^{(l-1)} + \mathbf{Z}). \quad (8)$$

Then enter the position by position feedforward network, as shown in Formula (9).

$$\mathbf{H}^{(l)} = \text{LayerNorm}(\tilde{\mathbf{H}}^{(l)} + \text{FFN}(\tilde{\mathbf{H}}^{(l)})), \text{FFN}(x) = \max(0, x\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2. \quad (9)$$

4. The Whole Word Masking policy, RoBERTa-wwm-ext, randomly masks complete words in the pre training phase, as shown in Formula (10).

$$x_i = \begin{cases} [\text{MASK}], & 80\% \\ \text{Random token}, & 10\% \\ x_i, & 10\%. \end{cases} \quad i \in M. \quad (10)$$

Where, M is the complete set of index words selected. The strategy is in the loss function, as shown in Formula (11).

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in M} \log P(x_i | \mathbf{H}^{(L)}). \quad (11)$$

Where, $H^{(L)}$ is the 12th layer output, and $P()$ is the prediction probability of Softmax. This strategy forces the model to restore the whole word, and improves the overall modeling ability of medical compound words.

- TF-IDF Features Extraction.

In addition to deep semantic representation, this paper uses word frequency inverse document frequency (TF-IDF) to capture the significance of medical keywords, and additionally constructs category level prototypes to provide a priori constraint for long tail categories.

At the sentence level, TF-IDF constructs a vocabulary for all queries in the training set and retains 200 feature words with the highest amount of information (max_features=200, the amount of information is sorted according to the χ^2 score to reduce noise). Each query d is first normalized to Unicode and lowercase in English. After deleting the stop words and punctuation, it is divided into token sequences according to “space + Chinese characters”. This dimensionality control (max_features=200 and shared linear projection to 768-d) ensures that TF-IDF vectors remain compact and computationally efficient while preserving critical keyword signals for ultra-short medical queries. Let the vocabulary size be V , and the frequency of the t th word in sample d is shown in formula (12).

$$\text{tf}_{t,d} = \frac{n(t,d)}{\sum_t' n(t',d)} \quad (12)$$

If the document frequency is $\text{df}_t = |\{d: n(t,d) > 0\}|$, then the reverse document frequency after smoothing is shown in formula (13).

$$\text{idf}_t = \log \left(\frac{N+1}{\text{df}_t+1} \right) + 1 \quad (13)$$

Where, N is the total number of training corpus documents. The final weight is shown in Formula (14).

$$\text{tfidf}_{t,d} = \text{tf}_{t,d} \times \text{idf}_t. \quad (14)$$

The sparse vector V_{sent} is normalized to L2, and then projected to the deep semantic space through the shared linear layer W_{proj} , to form sentence level keyword expression.

Class level TF-IDF, in order to inject a global prior, first divide the sample into 11 sets D_k according to the true label of the training set. The prototype vector definition of each type of k is shown in Formula (15).

$$\bar{\mathbf{v}}_k = \frac{1}{|D_k|} \sum_{d \in D_k} \mathbf{v}_{sent}^{(d)}, \quad k = 1 \dots 11 \quad (15)$$

Similarly, L2 normalization and linear mapping were performed, and it was found that real labels could not be directly used in the c_k inference phase, so soft prototype fusion was adopted, as shown in Formula (16).

$$\mathbf{v}_{class} = \sum_{k=1}^{11} p_k \mathbf{c}_k, \quad p_k = \Pr(y = k | q; \theta_{prev}) \quad (16)$$

The p_k comes from the predicted distribution of the current query in the previous epoch model. This strategy is equivalent to making expectations on the category prototype, which can avoid information leakage and iteratively refine the prior.

Through these two TF-IDF representations, the model enhances the sensitivity to medical keywords and category characteristics while keeping the parameters lightweight, and provides high-quality, low-noise lexical prior for the attention fusion layer.

Experiments

Problem definition and dataset description

In medical triage, accurate intent classification of patient queries significantly enhances operational efficiency. However, the profound complexity of medical domain knowledge and the inherent ambiguity/diversity of user expressions pose substantial challenges for precise categorization. Medical inquiries span a broad spectrum of scenarios, including disease diagnosis, etiology analysis, therapeutic regimens, and hospital guidance, necessitating classification models with both strong generalization capacity and fine-grained discriminative power. This study addresses these challenges within the context of mobile medical search, where queries are typically extremely short, use colloquial terminology, and exhibit severe class imbalance. The model must jointly capture contextual semantics and identify critical medical terminology to achieve high-precision multi-class intent determination. The experiments are conducted on the KUAKE-QIC dataset, sourced from the Chinese Biomedical Language Understanding Evaluation (CBLUE) benchmark maintained by Alibaba Cloud's Tianchi platform (aliyun.com). The dataset comprises 6,931 training samples and 1,955 validation samples, annotated across 11 intent categories: diagnosis (As detailed in Table 4), cause analysis (cause), therapeutic methods (therapy), hospitalization guidance (advice), laboratory metric interpretation (metric), disease description (diseaseDesc), prognostic outcomes (outcome), clinical precautions (caution), therapeutic efficacy (effect), medical costs (cost), and an other category. As detailed in Table 5 training set class distribution, therapeutic methods and "other" categories dominate the dataset, while critical classes such as cause analysis and metric interpretation are severely underrepresented. This task thus confronts dual challenges: class imbalance and sparse contextual information in ultra-short text queries.

Evaluation metrics

In this experiment, Accuracy, Macro-Average Precision, Macro-Average Recall and Macro-Average F1 Score, Micro-Average F1 Score and Confusion Matrix were used to evaluate the experimental results.

Accuracy refers to the ratio of the number of samples correctly predicted by the model to the total number of samples, as shown in Eq. (17).

$$A_c = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

Where, TP represents the number of samples that are actually positive and correctly predicted as positive. TN represents the number of samples that are actually negative and correctly predicted as negative. FP represents the

Number	Medical issues	Type description	typical example
0	Diagnosis	Known symptoms, determine possible causes	What's with the recent weakness in the morning?
1	Cause analysis	Known diseases, explain why they occur	How does nasopharyngeal cancer occur?
2	Therapeutic methods	Known diseases/symptoms, give options for treatment or relief	What is the best medicine for a cold headache?
3	Hospitalization guidance	Known symptoms/diseases, advice on where to seek medical attention	What tests should I get for diabetes?
4	Laboratory metric interpretation	Interpretation of numerical ranges for height/blood pressure and other test results	What does a high blood count of ultrasensitive C-reactive protein indicate?
5	Disease description	Disease attributes	Vitiligo can be cured
6	Prognostic outcomes	Diseases/symptoms/medicines/test items/food hazards	What happens when hepatitis B is left untreated
7	Clinical precautions	Things to look out for in patients	What to look for in asthma
8	Therapeutic efficacy	Benefits of food/drugs, efficacy/effects/side effects	Effects of Wu Ji Bai Feng Wan
9	Medical costs	Costs of diseases/surgeries/drugs/examinations/	How much do you want for Dimethyldipicarb?
10	Other	Unexplained needs	What are the five categories of blood tests?

Table 4. Data description table.

Category	Count	Percentage
Therapeutic methods	1,750	25.25%
Other	1,617	23.33%
Diagnosis	877	12.65%
Clinical precautions	650	9.38%
Disease description	594	8.57%
Hospitalization guidance	371	5.35%
Therapeutic efficacy	370	5.34%
Prognostic outcomes	235	3.39%
Medical costs	177	2.55%
Cause analysis	153	2.21%
Laboratory metric interpretation	137	1.98%

Table 5. Class distribution in training Set.

number of samples that are actually negative but incorrectly predicted as positive. FN represents the number of samples that are actually positive but incorrectly predicted as negative.

Macro-average Precision is the average of the precision rates for each category, without taking into account the influence of the sample quantity of each category. Specifically, it is as shown in Formula (18).

$$P_{Ma} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \quad (18)$$

Where, N represents the total number of categories. TP_i denotes the true positive count of the i category. FP_i indicates the false positive count of the i category.

Macro-average Recall is the average of recall rates for each category, without taking into account the influence of the number of samples in each category. Specifically, it is as shown in Formula (19).

$$R_{Ma} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (19)$$

Where, N represents the total number of categories. TP_i denotes the true positive count of the i category. FN_i indicates the false positive count of the i category.

Macro-average F1 Score is the average of F1 scores for each category, without taking into account the influence of the sample quantity of each category. Specifically, it is as shown in Formula (20).

$$F1_{Ma} = \frac{1}{N} \sum_{i=1}^N \frac{2 \times Pr_i \times Re_i}{Pr_i + Re_i} \quad (20)$$

Where, N represents the total number of categories. Pr_i denotes the precision rate of the i category. Re_i indicates the recall rate of the i category.

Parameter	Value
Backbone Network	RoBERTa-wwm-ext
Maximum Sequence Length	256
Batch Size	16
Optimizer	AdamW
Initial Learning Rate	2.00×10^{-5}
Learning Rate Schedule	Linear decay with 10% warm-up steps
TF-IDF Feature Dimension	200
Projected Feature Dimension	768
Dropout Rate	0.2
Loss Function	Cross-Entropy Loss
Total Training Epochs	10

Table 6. Model training parameter setting Table.

Epoch	A_c	P_{Ma}	R_{Ma}	$F1_{Ma}$	$F1_{Mi}$	ROC AUC	Av Loss
1	0.788	0.621	0.643	0.624	0.788	0.9475	1.853
2	0.794	0.74	0.811	0.76	0.794	0.9729	0.819
3	0.816	0.78	0.798	0.782	0.816	0.9733	0.504
4	0.804	0.758	0.813	0.778	0.804	0.9712	0.347
5	0.819	0.775	0.811	0.79	0.819	0.9714	0.263
6	0.822	0.772	0.829	0.79	0.822	0.9695	0.184
7	0.824	0.791	0.815	0.8	0.824	0.9662	0.141
8	0.793	0.743	0.814	0.761	0.793	0.9573	0.137
9	0.802	0.745	0.793	0.765	0.802	0.9631	0.095
10	0.812	0.753	0.804	0.773	0.812	0.9577	0.079

Table 7. Index Change.

The Micro-average F1 Score is calculated by first determining the overall precision and recall rates, and then computing their F1 scores. Specifically, it is as shown in Formula (21).

$$F1_{Mi} = \frac{2 \times Pr_{mi} \times Re_{mi}}{Pr_{mi} + Re_{mi}} \tag{21}$$

Where, Pr_{mi} represents the micro-averaged precision rate. Re_{mi} represents the micro-averaged recall rate.

Implementation details

This experiment was carried out in the NVIDIA RTX 5090 GPU (32GB video memory) environment, using Python 3.10 and PyTorch 2.1. The model training parameter settings are shown in Table 6.

10 epochs were performed in the model training. Continuous monitoring of training loss and validation set Macro-F1 indicators during training. The change trend is shown in Fig. 3. The specific changes of A_c , P_{Ma} , R_{Ma} , $F1_{Ma}$, $F1_{Mi}$, ROC AUC, Av Loss evaluation indicators during each epoch training are shown in Table 7.

The training results in Fig. 2 show that the model achieves the best generalization performance in the seventh epoch, and the continuous training has not significantly improved since then. The final experimental report uses the seventh round model as the benchmark for evaluation and ablation experiments.

As shown in Table 7, the model demonstrated rapid improvement in evaluation metrics during initial training stages, particularly for Accuracy, Macro Precision, Macro Recall, Macro-F1, and Micro-F1, which achieved significant growth within the first 3 epochs. This indicates the model's efficient capture of core task features. With increasing training epochs, Macro-F1 reached its peak value of 0.800 at the 7th epoch, after which all metrics stabilized, suggesting the model entered a performance plateau. Notably, the ROC AUC metric peaked at 0.973 during the 3rd epoch and maintained high levels subsequently, demonstrating strong multi-class discrimination capability. The average loss exhibited consistent decline throughout training, indicating satisfactory convergence behavior. Collectively, the model achieved optimal performance at the 7th epoch, with subsequent loss reduction not translating to significant generalization improvement.

Results and analysis
Comparison with baselines

To systematically evaluate the effectiveness of the proposed methods, we compare the model proposed in this paper (recorded as My Model) with three types of baselines:

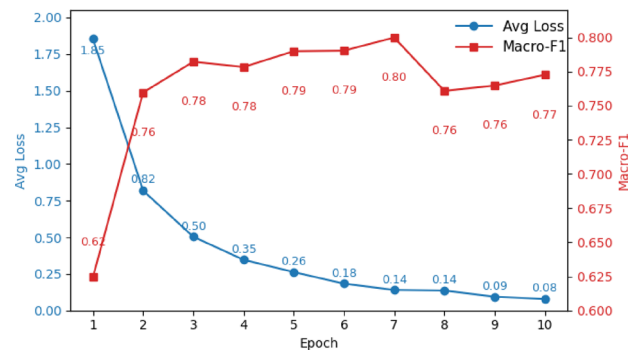


Fig. 2. Changes of Avg Loss and Macro-F1.

Model	Category	Feature representation & architecture	Key characteristics
TF-IDF + LR	Traditional	Sparse TF-IDF vectors → Logistic Regression	Classic text classification baseline; simple implementation with fast inference, but limited to word frequency features without contextual semantics.
Word2Vec + LR	Traditional	Pre-trained Word2Vec averaging → Logistic Regression	Dense distributed representations capture word similarity; sentence-level averaging loses sequential information, performance constrained for long texts.
BOW + LR	Traditional	Bag-of-words count vectors → Logistic Regression	User-friendly training and interpretability; suffers from high dimensionality, sparsity, and weak semantic representation.
LR	Traditional	1-gram counts → Logistic Regression	Similar to BOW, commonly used as machine learning course baseline.
SVM	Traditional	1-gram counts → Linear SVM	Effective for high-dimensional sparse features; sensitive to hyperparameters, moderate robustness to class imbalance.
TextCNN	Traditional	Random/pre-trained embeddings → Multi-scale 1D CNN → Max-pooling	Captures local n-gram patterns; shallow architecture with limited context window.
BERT	Deep Semantic	Transformer pre-trained LM (12 L, 768 d)	End-to-end fine-tuning; strong contextual modeling capability; high parameter count with significant inference cost.
BERT + CNN	Deep Semantic	Frozen/fine-tuned BERT → CNN classification head	Enhances local pattern recognition through convolutional aggregation; marginal parameter increase.
BERT + LR	Deep Semantic	BERT [CLS] vector → Logistic Regression	Fast training with good interpretability; performance constrained by linear decision boundary.
BERT + SVM	Deep Semantic	BERT average/[CLS] vector → Linear SVM	Projects deep semantics to hard-margin classifier; demonstrates certain advantages for minority classes.
BERT + RF	Deep Semantic	BERT embeddings → Random Forest	Non-linear ensemble classification captures complex boundaries; high storage overhead for high-dimensional features.
RoBERTa	Deep Semantic	RoBERTa-base (enhanced pre-training + dynamic masking)	Exhibits faster convergence and marginal accuracy improvement over BERT.
BERT + LDA + LR	Hybrid	BERT [CLS] vector + LDA topic distribution → Logistic Regression	Combines global topic priors with deep semantics; complex feature engineering and linear classifier limits performance ceiling.
My Model	Hybrid	(1) RoBERTa-wwm-ext [CLS] representation (2) Sentence-level 200-d TF-IDF projection (3) Class-centroid TF-IDF projection → Attention-weighted fusion → MLP classification	Adaptive deep-shallow feature complementarity; explicit class centroids enhance long-tail recognition; significantly improves macro-average precision/ F1 while maintaining recall.

Table 8. Comparison of different models.

1. Traditional model. Including TF-IDF + LR, Word2V-ec + LR, Bow + LR, LR, SVM and TextCNN.
2. Deep semantic model. Include BERT²², BERT + C-NN⁴³, BERT + LR⁴⁴, BERT + SVM⁴⁵, BERT + RF⁴⁶, RoBERTa⁴⁷.
3. Hybrid model. Including BERT + LDA + LR⁴ and My Model.

The specific characteristics of each model are shown in Table 8 Comparison of Different Models. The comparison results are shown in Table 9.

It can be seen from the results in Table 9 that the traditional shallow model, because it only uses the word frequency or static word vector, lacks the ability of context modeling, and all five indicators are lower than 0.40, Macro-F1 is even less than 0.07, which indicates that in the task facing 11 types of medical intentions, which is highly semantic dependent, simple sparse or average vectors are difficult to reveal the complex relationship between keywords and contexts; SVM improves the macro average precision (0.251) with the help of hard spaced hyperplane, but the recall is still limited (0.101), resulting in limited F1 improvement. After the introduction of TextCNN, the local n-gram semantics are captured to some extent, and the Macro Precision is improved to 0.213. However, the overall gain is still limited due to the length of the convolution window and the size of the corpus. After turning to the deep semantic model, there is a qualitative leap in performance: pure BERT fine-tuning

Model	A_c	P_{Ma}	R_{Ma}	$F1_{Ma}$	$F1_{Mi}$
TF-IDF + LR	0.359 ± 0.012	0.154 ± 0.010	0.099 ± 0.012	0.065 ± 0.008	0.359 ± 0.015
Word2Vec + LR	0.346 ± 0.015	0.031 ± 0.005	0.091 ± 0.010	0.047 ± 0.006	0.346 ± 0.018
Bow + LR	0.360 ± 0.013	0.156 ± 0.012	0.099 ± 0.011	0.065 ± 0.007	0.360 ± 0.016
LR	0.362 ± 0.011	0.160 ± 0.014	0.101 ± 0.009	0.066 ± 0.005	0.362 ± 0.013
SVM	0.361 ± 0.010	0.251 ± 0.020	0.101 ± 0.004	0.067 ± 0.008	0.361 ± 0.012
TextCNN	0.347 ± 0.014	0.213 ± 0.018	0.094 ± 0.013	0.053 ± 0.003	0.347 ± 0.017
BERT	0.802 ± 0.008	0.744 ± 0.005	0.792 ± 0.006	0.764 ± 0.004	0.802 ± 0.009
BERT + CNN	0.803 ± 0.007	0.745 ± 0.004	0.816 ± 0.005	0.769 ± 0.003	0.803 ± 0.008
BERT + LR	0.801 ± 0.009	0.741 ± 0.006	0.816 ± 0.004	0.767 ± 0.002	0.801 ± 0.010
BERT + SVM	0.807 ± 0.006	0.749 ± 0.003	0.797 ± 0.005	0.768 ± 0.002	0.807 ± 0.007
BERT + RF	0.813 ± 0.005	0.787 ± 0.002	0.781 ± 0.003	0.774 ± 0.002	0.813 ± 0.004
RoBERTa	0.811 ± 0.007	0.755 ± 0.004	0.815 ± 0.005	0.777 ± 0.003	0.811 ± 0.008
BERT + LDA + LR	0.764 ± 0.012	0.562 ± 0.015	0.561 ± 0.010	0.552 ± 0.008	0.764 ± 0.014
My Model	0.824 ± 0.008	0.791 ± 0.005	0.815 ± 0.004	0.800 ± 0.003	0.824 ± 0.006

Table 9. Evaluation of model Results.

Variant	Composition	Description
Full	R + S + C + Attn	Full model
Remove C	R + S + Attn	Removes class-centroid branch (C)
Remove S	R + C + Attn	Removes sentence-level TF-IDF branch (S)
Remove TF-IDF	R + Attn	Semantics-only baseline using only RoBERTa (R)
Concat	R + S + C	Naive concatenation of features without attention mechanism

Table 10. Description of ablation test.

can increase Accuracy and Macro-F1 to 0.802 and 0.764 respectively, indicating that context semantics plays a decisive role in intention classification; However, the reuse of BERT vectors by different subsequent classifiers did not bring about a costly global improvement. BERT + CNN and BERT + LR sacrificed their accuracy while the Macro Recall reached 0.816. Finally, F1 was slightly lower; Although BERT + RF is ahead of Macro Precision (0.787), its overall advantage is not obvious because of its recall to 0.781. After further using RoBERTa, which is more fully pre trained, all indicators are slightly better than BERT, but the increase is limited, indicating that the bottleneck is approaching only by deep semantic branches. In the mixed feature route, simple superimposed topic distribution (BERT + LDA + LR) brings serious dimensional disaster, precision recall declines both, and verification of undifferentiated splicing will dilute the discrimination information. In contrast, the deep shallow feature attention fusion model proposed in this paper significantly improves the macro average accuracy to 0.791 while maintaining a high recall rate (0.815) through the dynamic weighting of sentence level TF-IDF, category centroid TF-IDF, and RoBERTa semantic vectors. Finally, Macro-F1 reaches 0.800, Accuracy and Micro-F1 both reach 0.824, 2.4 pp, 3.6 pp, and 1.3 pp higher than the strongest single-mode RoBERTa, respectively. This shows that quasi prior and keyword clues can effectively enhance the ability to distinguish long tail categories, while the attention mechanism avoids information redundancy, thus achieving an optimal balance on multiple indicators.

Ablation study

In order to verify the contribution of each component module to the overall performance, this paper designs five groups of control tests around RoBERTa_wwm_ext semantic branch (R), sentence level TF IDF branch (S), category centroid TF IDF branch (C), and attention fusion layer (Attn), as shown in Table 10.

It can be seen from the ablation experiment results in Table 11 that the complete model (Full) maintains the highest level in five indicators, of which Macro-F1 reaches 0.800, indicating that sentence level TF-IDF, category centroid TF-IDF and attention fusion have the best synergistic effect. When the centroid branch of the category is removed (C is removed), Macro Precision decreases from 0.791 to 0.773, and Macro-F1 decreases by 1.1 pp, indicating that the category prototype vector provides a key priori for identifying long tail categories; The removal of sentence level TF-IDF (removal of S) causes a loss of precision of similar magnitude (Macro-F1 decreases by 1.4 pp), indicating that keyword level sparse features are also important for capturing local semantics. When the two shallow features (TF-IDF, equivalent to pure RoBERTa) are further removed at the same time, Macro Precision drops to 0.755 and Macro-F1 drops to 0.777, 2.3 pp less than the complete model, highlighting the overall value of deep shallow complementarity. Replacing attention fusion with simple concat retains all information, but Macro-F1 is only 0.786, 1.4 pp lower than Full, and its precision and recall decline synchronously, indicating that static splicing is difficult to balance the contributions of various branches, and the attention mechanism can dynamically allocate weights according to samples to avoid information dilution.

Variant	A_c	P_{Ma}	R_{Ma}	$F1_{Ma}$	$F1_{Mi}$
Full	0.824 ± 0.008	0.791 ± 0.005	0.815 ± 0.004	0.800 ± 0.003	0.824 ± 0.006
Remove C	0.818 ± 0.009	0.773 ± 0.006	0.811 ± 0.005	0.789 ± 0.004	0.818 ± 0.008
Remove S	0.814 ± 0.010	0.770 ± 0.007	0.806 ± 0.006	0.786 ± 0.005	0.814 ± 0.009
Remove TF-IDF	0.811 ± 0.007	0.755 ± 0.004	0.815 ± 0.005	0.777 ± 0.003	0.811 ± 0.008
Concat	0.812 ± 0.007	0.769 ± 0.005	0.805 ± 0.004	0.786 ± 0.003	0.812 ± 0.006

Table 11. Results of ablation test.

Factor	Setting	$F1_{Ma}$	A_c	Time(s)
Attention	Additive	0.794 ± 0.007	0.813 ± 0.005	171
	Dot	0.799 ± 0.002	0.821 ± 0.004	175
	MH(4)	0.799 ± 0.004	0.820 ± 0.007	188
	MH(8)	0.795 ± 0.002	0.818 ± 0.006	188
TF-IDF max features	100	0.787 ± 0.006	0.818 ± 0.004	205
	200	0.799 ± 0.007	0.823 ± 0.005	171
	400	0.786 ± 0.004	0.815 ± 0.002	195
	800	0.788 ± 0.008	0.817 ± 0.004	186
Max input length	64	0.791 ± 0.005	0.815 ± 0.005	123
	128	0.793 ± 0.007	0.819 ± 0.005	187
	256	0.800 ± 0.003	0.824 ± 0.008	265
	512	0.798 ± 0.004	0.815 ± 0.005	573
Dropout	0.1	0.796 ± 0.008	0.816 ± 0.010	172
	0.2	0.799 ± 0.006	0.824 ± 0.007	169
	0.3	0.794 ± 0.007	0.813 ± 0.005	175

Table 12. Hyperparameter sensitivity.

It can be seen from comprehensive comparison that the two TF-IDF branches each bring about more than 1 pp of Macro-F1 improvement, and the attention fusion is more than splicing to obtain about 1.4 pp of gain. The superposition of the three makes the model significantly improve accuracy and overall Macro-F1 without sacrificing recall, which verifies the effectiveness and necessity of the deep shallow feature complementarity and adaptive fusion strategy.

Effectiveness of Shallow Semantics. Sentence-level TF-IDF emphasizes symptom- and drug-related keywords that deep contextual encoders may smooth out, sharpening decision boundaries for short or ambiguous queries. Category-centroid TF-IDF introduces global lexical priors, acting as prototype anchors that stabilize decision boundaries for minority classes. Removing either shallow branch results in noticeable drops in Macro-F1 (−1.2 to −1.5 pp), while removing both reduces performance by 2.3 pp. These results confirm that deep-shallow complementarity effectively balances precision and recall, particularly under long-tail distributions.

Hyperparameter tuning and sensitivity

From the Hyperparameter sensitivity in Table 12. We conducted one-at-a-time sweeps with five random seeds per setting (10 epochs, early stopping), fixing the projection dimension at 768 and holding other factors at their baseline (batch size 16, AdamW, lr = 2e-5, warmup = 10%, weighted cross-entropy). Results are reported as mean ± std on the dev set. As summarized in Table 12, dot-product attention and multi-head(4) yield the highest Macro-F1 (≈ 0.799), slightly outperforming additive attention, with comparable efficiency; we therefore adopt dot attention as default. The model is most sensitive to maximum input length and TF-IDF vocabulary size: performance peaks at max_len = 256 (Macro-F1 = 0.800 ± 0.003; Ac = 0.824 ± 0.008) and tfidf_max_features = 200 (Macro-F1 = 0.799 ± 0.007), while larger vocabularies (≥ 400) degrade Macro-F1 on ultra-short queries, likely due to sparsity noise. Dropout = 0.2 offers the best generalization among tested values. Overall, the best configuration is attn = dot, tfidf_max_features = 200, max_len = 256, dropout = 0.2, delivering Macro-F1 ≈ 0.800 and Accuracy ≈ 0.824 with moderate training cost.

Error analysis

In order to further understand the performance of the model at the fine-grained level, we obtained the complete Classification Report as shown in Table 13 and the corresponding confusion matrix as shown in Fig. 4. The Report lists the precision, recall rate and F1 score of 11 intention categories; The confusion matrix visually shows the corresponding relationship between the real tag and the predicted tag.

From the category level indicators in Table 13, we can see that the model shows obvious head tail differentiation between different intentions: the main categories with large sample size and highly fixed semantic structure (such as “Therapy” and “Diagnosis”) and the standardized categories containing significant keywords (such as

Label	Precision	Recall	F1-score	Support
Other	0.68	0.6	0.64	395
Effect	0.77	0.82	0.79	28
Cost	1	0.96	0.98	50
Outcome	0.55	0.69	0.61	45
Advice	0.84	0.98	0.9	134
Metric	0.79	0.69	0.73	32
Therapy	0.93	0.91	0.92	676
Caution	0.72	0.9	0.8	120
DiseaseDesc	0.78	0.75	0.76	158
Cause	0.79	0.79	0.79	29
Diagnosis	0.85	0.88	0.86	288

Table 13. Classification Report.

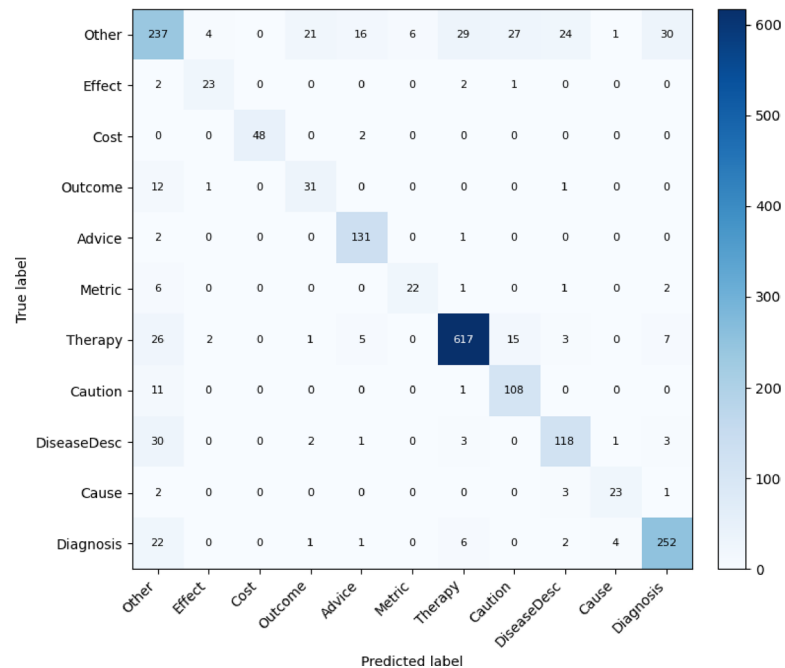


Fig. 3. Confusion matrix.

“Cost”) have achieved F1 score close to or more than 0.90, which shows the robust performance of deep semantic representation and shallow keyword features; On the contrary, categories with fuzzy semantic boundaries or strong emotional color (such as “Outcome” and “others”) become the main source of error. Although the recall rate of “Outcome” has increased to 0.69, the precision is only 0.55, which indicates that the model is still difficult to distinguish between emotional inquiry and disease description, while F1 score of “others” category 0.64 reflects a lot of noise brought by open semantic space. The model also showed good generalization ability in the long tail category. Only 28 samples’ “Effect” achieved a F1 score of 0.79 based on the reinforcement of sentence level and category centroid TF-IDF, and “Cause” and “Metric” also reached 0.79 and 0.73 respectively, proving that shallow features are of great value to low resource categories. In general, deep shallow feature attention fusion has effectively improved the accuracy and recall of most categories, especially in small sample scenarios. However, for categories with semantic overlap or rich emotional expression, hierarchical tags, external medical knowledge or confidence screening mechanisms need to be further introduced to further reduce misjudgments and improve overall macro performance.

It can be seen from the confusion matrix in Fig. 3 that the model maintains a strong discriminant ability in most categories, and the counts of “Therapy”, “Cost”, “Advice” and other trunk categories on the diagonal are the most concentrated, almost without being misclassified, which verifies the high reliability of deep semantics and key word features. However, the error still presents a typical long tail migration phenomenon. On the one hand, the “Other” category spreads most seriously outward, with 108 samples being wrongly cast into other columns, indicating that open semantic tags still act as noise collectors in practical applications. On the other hand, categories with emotional or ambiguous semantic boundaries interfere with each other, in which “Outcome”

was misjudged as “Other” 12 times and “DiseaseDesc” 1 time, reflecting the overlap between emotional inquiry and symptom description. There were 15 mutual errors between the precautions and the treatment plan, because they shared keywords such as “taking medicine, avoiding eating”. In addition, the “Effect” and “Cause” with the smallest sample size still maintain a high recall, which proves the reinforcement effect of sentence level and category centroid TF-IDF on the long tail category. In general, misjudgments mainly focus on the intersection of “other → multi category” and a few semantically similar categories. In the next step, these errors can be further compressed by subdividing “other” tags.

Discussion

This work examined a hybrid architecture that fuses deep contextual representations (RoBERTa-wwm-ext) with sentence-level TF-IDF and class-centroid TF-IDF via attention. The comparative study (Table 9) shows a clear advantage over traditional shallow models and single-branch deep encoders, with the proposed model attaining Accuracy = 0.824 and Macro-F1 = 0.800 on KUAKE-QIC. The ablation analysis (Table 11) indicates that both lexical branches contribute complementary gains (>1 pp each in Macro-F1) and that adaptive attention is superior to naïve concatenation (+1.4 pp), underscoring the necessity of (i) explicit lexical priors and (ii) sample-wise feature re-weighting. Hyperparameter sensitivity (Table 12) further suggests that performance is most responsive to max input length and TF-IDF vocabulary size; with ultra-short medical queries, max_len = 256 and tfidf_max_features = 200 strike the best balance between semantic coverage and sparsity control, while larger vocabularies (≥400) introduce noise that depresses Macro-F1. Among attention variants, dot-product and multi-head (4) yield the strongest Macro-F1 (0.799) with comparable training time, supporting dot attention as a practical default.

At the label level (Table 13; Fig. 3), the model delivers near-ceiling performance on head or keyword-salient classes (e.g., Therapy, Diagnosis, Cost), confirming that deep semantics and keyword salience are jointly captured. The residual errors concentrate in semantically diffuse or affect-laden categories (e.g., Outcome, Other), where boundaries overlap with DiseaseDesc and sentiment-like intents. We also observe “long-tail migration” in the confusion matrix: Other disperses widely, acting as a sink for open-set queries; nevertheless, minority classes (Effect, Cause, Metric) benefit from the centroid prior and sentence-level TF-IDF, maintaining competitive F1 despite limited support. Based on the above analysis, further improvements can be made: (1) label space refinement (sub-typing Other), (2) knowledge-augmented features (medical ontologies, synonym maps) to reduce ambiguity, and (3) confidence-based rejection/handoff to human triage to mitigate high-risk misclassifications.

Generalisability and deployment. The method targets ultra-short Chinese medical queries with class imbalance. While the fusion design is model-agnostic, transferability to other domains/languages may depend on domain lexicons and centroid stability. For real-world use, the footprint remains moderate (single RoBERTa encoder + two lightweight projections + small MLP), and attention adds minimal latency; measured training/inference times in Table 12 suggest feasibility for online intent routing.

Limitations and future directions. (1) Single-dataset evaluation: results may overestimate in-domain performance; cross-dataset validation (e.g., other CBLUE tasks or real-world logs) is a priority. (2) Open-set and ambiguity: Other and Outcome remain challenging; hierarchical tagging and ontology-aware disambiguation are promising. (3) LLM comparison: when LLMs are available, few-shot prompting can be competitive; however, our hybrid remains attractive for efficiency, stability, and controllability. Future work will integrate structured medical knowledge (UMLS/CHIP ontologies), explore sequence-aware variants for dialogic contexts, and incorporate confidence calibration with selective prediction.

Conclusion

On the KUAKE-QIC benchmark, the proposed RoBERTa-wwm-ext + sentence TF-IDF + category-centric TF-IDF framework with attentive fusion delivers Macro-F1 = 0.800 and Accuracy = 0.824, validating the effectiveness of deep–shallow complementarity and adaptive weighting.

- (1) Compared with the RoBERTa-only baseline, our model improves Macro-F1 by 2.3% points and Accuracy by 1.3% points, without substantial complexity increase.
- (2) Ablation studies show that each TF-IDF branch contributes ≈ 1 pp Macro-F1 individually, and attentive fusion provides an additional ≈ 1.4 pp over static concatenation—significantly boosting macro-precision without sacrificing recall, especially for long-tail categories.
- (3) Category-level analysis and the confusion matrix indicate strong performance on structurally coherent, keyword-rich classes, while TF-IDF centroid priors help rare classes maintain high F1, confirming the generalization of category-specific representations under class imbalance.

Overall, the approach is accurate, parameter-efficient, and deployment-friendly, providing a practical recipe for intent classification in ultra-short medical queries.

Data availability

Publicly available datasets were analyzed in this study. This data can be found here: <https://github.com/CBLUE/benchmark/CBLUE>.

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Author contributions

Zejian Liang spearheaded data preprocessing, model training, result validation, visualization and initial draft preparation; Yunxiang Zhao led research design, experimental framework development and algorithm optimization; Haiwen Xu conducted theoretical derivation, formula validation, literature review, and ethics compliance; Hong Huang managed experimental environment setup, code debugging, technical troubleshooting, and supplementary experiments; Luning Chen oversaw project supervision, funding acquisition, manuscript revision, and final approval. All authors contributed to revisions, approved the final version, and agree to be accountable for the work's integrity.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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